

Apply filters to SQL queries

Project description

As a security professional, my task is to identify and investigate potential security risks by querying the organization's log_in_attempts and employees databases. I will use SQL queries with specific filters to isolate suspicious activities and gather relevant employee data, focusing on login anomalies and departmental information.

Retrieve after hours failed login attempts

I'll query the log_in_attempts table to look into a possible security incident that happened after work hours. The purpose of this query is to identify all login attempts that took place after 18:00 (6:00 PM) and failed (where success is 0).

```
SELECT *  
FROM log_in_attempts  
WHERE success = 0 AND TIME(login_time) > '18:00:00';
```

Retrieve login attempts on specific dates

I must examine all login activity from May 8 and May 9, 2022, in order to assist an investigation into a particular incident. All entries in the log_in_attempts table that correspond to those two dates will be chosen by this query.

```
SELECT *  
FROM log_in_attempts  
WHERE login_date IN ('2022-05-08', '2022-05-09');
```

Retrieve login attempts outside of Mexico

Suspicious activity that did not come from Mexico has been found by our staff. I will filter the log_in_attempts dataset to identify all entries where the nation field does not start with "MEX" in order to isolate these occurrences. This method effectively eliminates "MEX" and "MEXICO" from the findings.

```
SELECT *  
FROM log_in_attempts  
WHERE country NOT LIKE 'MEX%';
```

Retrieve employees in Marketing

For a security upgrade, I must generate a list of employee computers. Employees in the "Marketing" department with offices in the "East" building are the target audience for this inquiry. All office names that begin with "East" are matched using the LIKE operator.

```
SELECT *  
FROM employees  
WHERE department = 'Marketing' AND office LIKE 'East%';
```

Retrieve employees in Finance or Sales

Machines in the Finance and Sales departments need a different security upgrade. All employee entries from the employees table where the department is either "Finance" or "Sales" will be retrieved by this query.

```
SELECT *  
FROM employees  
WHERE department IN ('Finance', 'Sales');
```

Retrieve all employees not in IT

A particular update has already been sent to the Information Technology department. I need a list of every employee outside of IT in order to distribute it around the entire organization. All personnel whose department is not "Information Technology" are selected by this query.

```
SELECT *  
FROM employees  
WHERE department != 'Information Technology';
```

Summary

I was able to retrieve particular datasets to aid in security investigations by using focused SQL WHERE clauses with operators like AND, IN, NOT LIKE, and !=. These queries identified certain employee groups by department, logins on particular dates, non-domestic login attempts, and after-hours failures. The security team can more successfully detect and address possible issues thanks to this data.